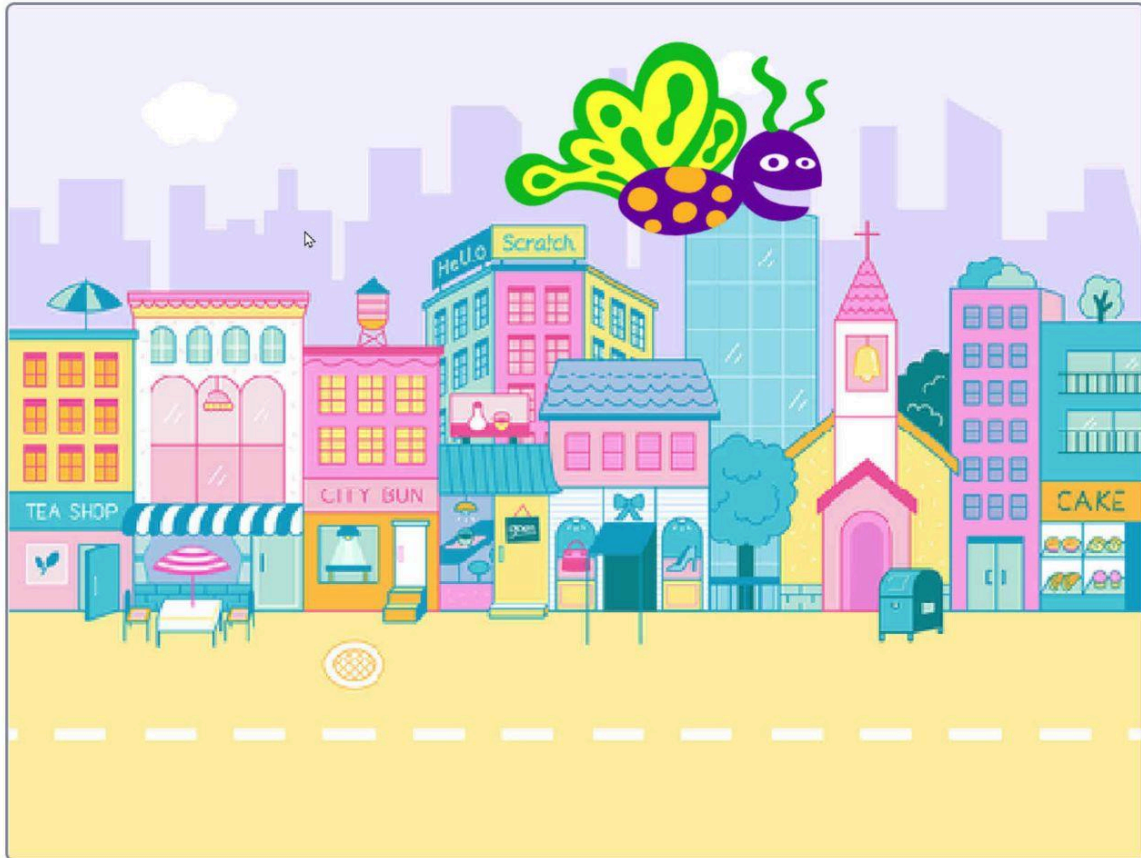


Unit 5

Looks & Sounds Workshop



Butterfly in the City

Week 5-6: Looks and Sounds Workshop

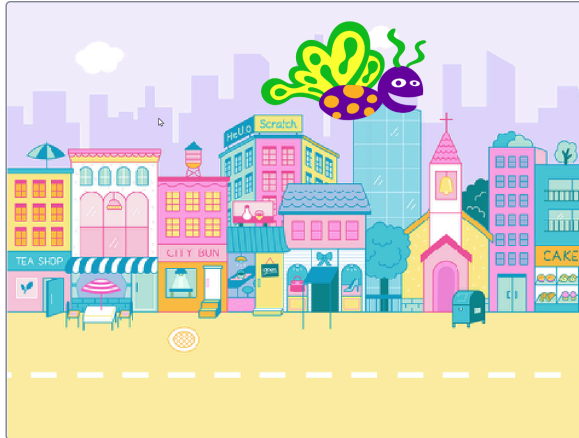
Save the file with a new name. Your new name should identify the project.

* Looks & Sounds * Butterfly in the City * Project 5

Workshop Summary

SCRATCH has amazing graphics and audio capabilities. The LOOKS menu controls the appearance of sprites and backdrops. The SOUNDS menu plays and records sounds for gameplay.

We will create a sampler of effects from the Looks and Sounds menu. The focus is on the fun and interest that Looks and Sounds can provide.



New Terms

Interface Terms

effects, vector, bitmap, curve, select, animation, costume

Code Block Terms

Sounds category, Looks category, loops, backdrop

Class Studio: <https://tinyurl.com/StchStudio01>

Video of Full Project: <https://vimeo.com/695494237>

This worksheet: <https://tinyurl.com/LooksWSPDF>

Starter Project:

<https://tinyurl.com/StchStarter05>

Full Project:

<https://scratch.mit.edu/projects/666680570>

Mini-Challenges in this Lesson

Click on the mini-challenge link to go directly to that section.

[Mini-Challenge 1: Use costumes for animation](#)

[Mini-Challenge 5: Use a sound](#)

[Mini-Challenge 2: Modify a costume](#)

[Mini-Challenge 6: Use a loop sound](#)

[Mini-Challenge 3: Set Graphic Effects](#)

[Mini-Challenge 7: Create a background change](#)

[Mini-Challenge 4: Change Graphic Effects](#)

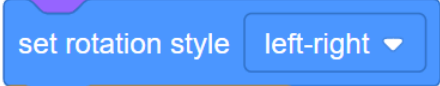
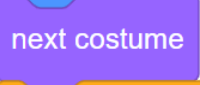
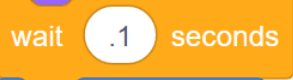
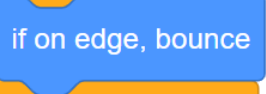
[Mini-Challenge 8: Change Soundtrack with BG](#)

Mini-Challenge 1: Use Costumes for Animation

Many Scratch sprites have more than one costume. A costume is a different view of the sprite. Some of the sprites have costumes that can be used for an animation. In animation, you switch between costumes to create the illusion that the sprite is walking, swimming, dancing, or whatever.

Procedure

1. Add a backdrop for your animation.
2. Add a new sprite that has multiple animation costumes.
3. Add a *when clicked* and *forever loop* to the sprite's code stack.
4. Inside the loop, add a next costume block.
 - a. Add a *move () steps* block.
 - b. Add a small *wait* such as 0.1 or 0.2 seconds.
5. Adjust the movement steps and wait time to make the sprite's movement seem natural.
6. If needed, add an *if on edge, bounce* block inside the loop.

Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BUTTERFLY sprite.
Looks	set size to ()		
Motion	set rotation style		
Control	forever		
Motion	move () steps		
Looks	next costume		
Control	wait (.1) seconds		
Motion	if on edge, bounce		
			

Mini-Challenge 2: Modify a costume

The costume tab is next to the code tab at the top of the page (above the Menu). This is where the poses or versions of each sprite are stored. On the costume tab, you can use paint tools to modify the appearance of the sprite’s costume.

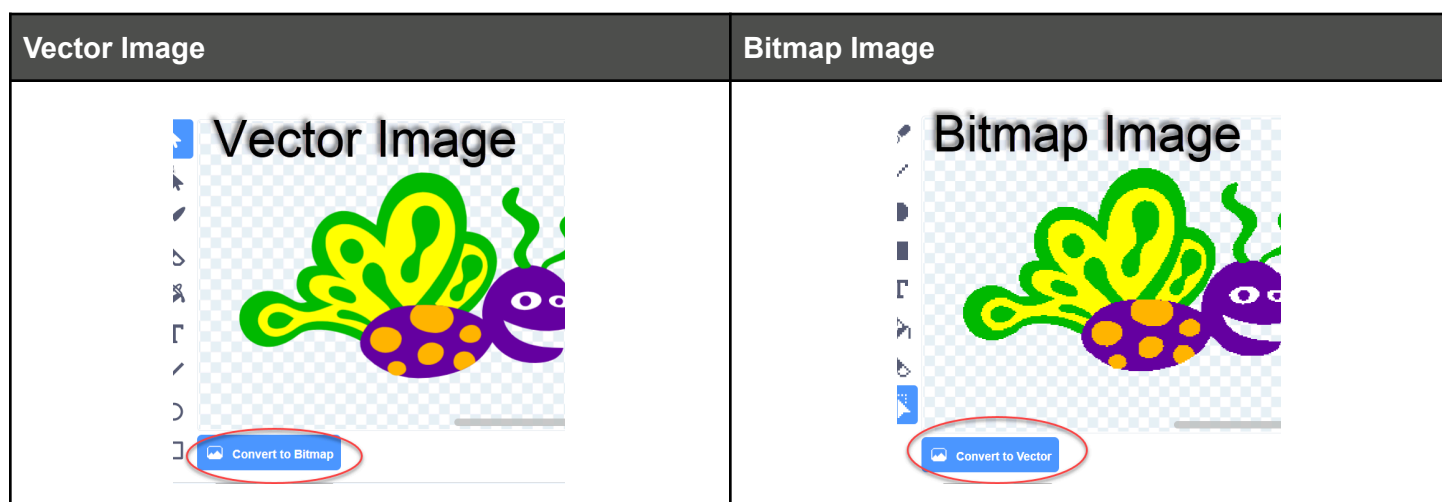
Procedure

- 1. Select a sprite on the stage.
- 2. Select the Costume tab.

Art Tool	Screenshot	Notes
The costumes tab is found between the code and sounds tabs.		

- 3. The available tools depend on the graphic format of the costume.
- 4. If the button at the bottom says **Convert to Bitmap**, the costume is a Vector image.
Vector images are the easiest kind of costume to modify because the image is made of many different shapes grouped.
Keep the costume as a vector when possible.
- 5. If the button at the bottom says **Convert to Vector**, the costume is a Bitmap image. This kind of costume doesn’t look as smooth and is harder to work with. Bitmap images are grouped into sections and are much more difficult to work with.

Converting a bitmap to a vector does not change this.



Procedure to Modify a Vector Graphic

1. Right-click the costume and duplicate it, then create a new file. This lets you make changes without ruining a good costume.

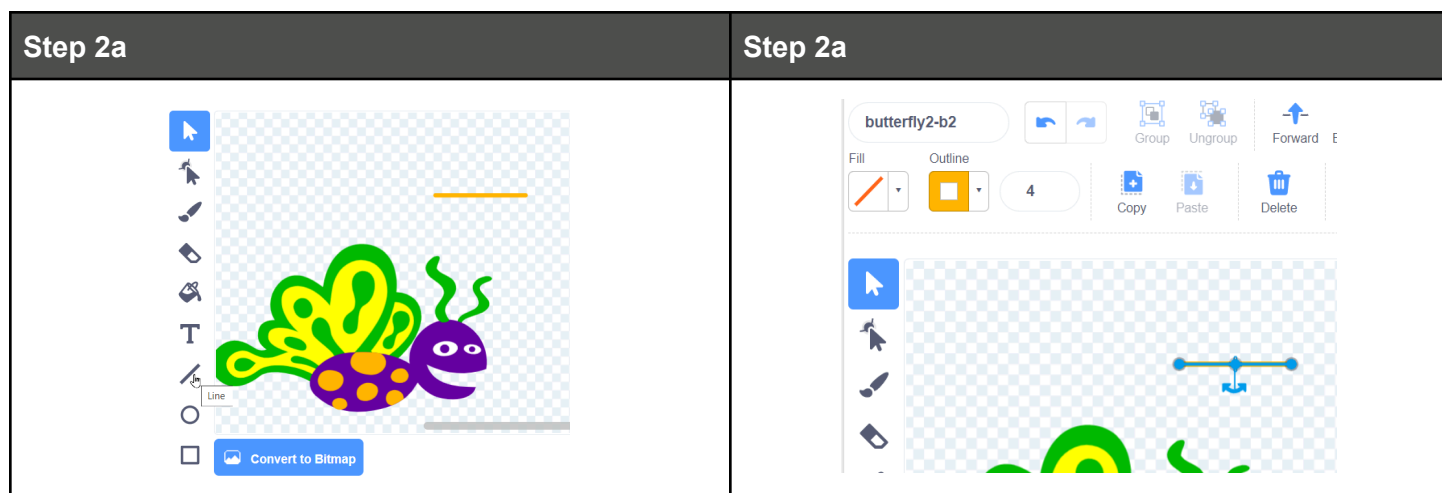
2. Draw a hat for the sprite.

For a simple hat, follow the instructions below:

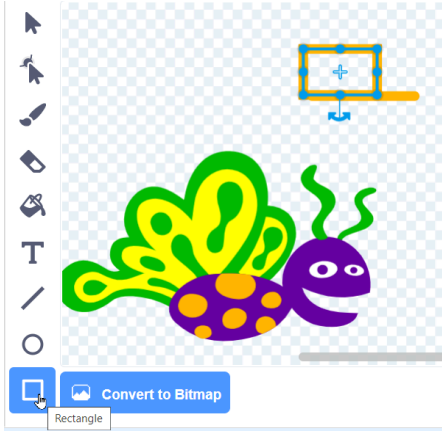
- a. Draw a line. Set the desired color and thickness.
- b. Draw a rectangle on top of the line. Set the desired fill color and outline size and color.

Pro-tip: Use the eyedropper tool to match a color already on the costume.

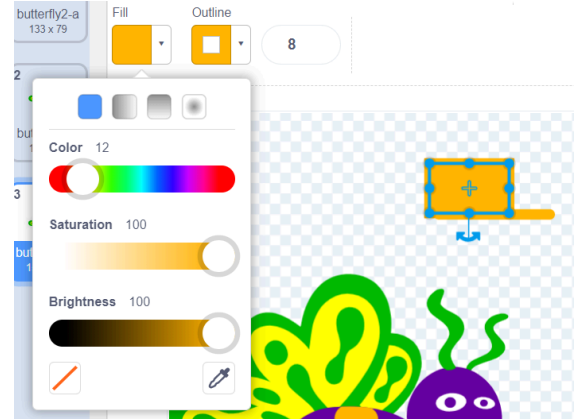
3. Use the **arrow tool** to select the line and rectangle.
4. Draw a box around the hat image to be modified, click the **“Group”** button.
5. Use the arrow tool to move the hat down to the sprite’s head. You can make it bigger or smaller and tilt the hat so it looks cool!
6. If you like the hat, you can select it with the arrow tool and copy it. Then you can paste it to the other costumes.
7. You can also use these tools to modify existing shapes on the sprite.



Step 2b



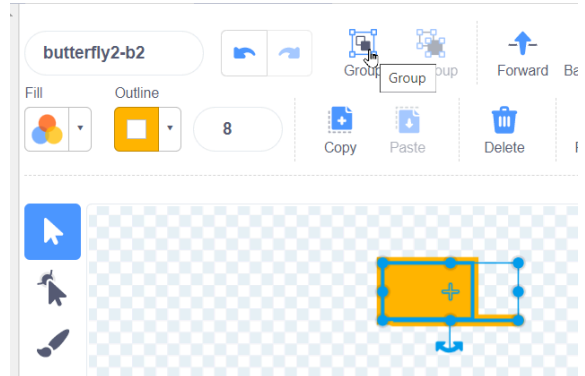
Step 2b



Step 3



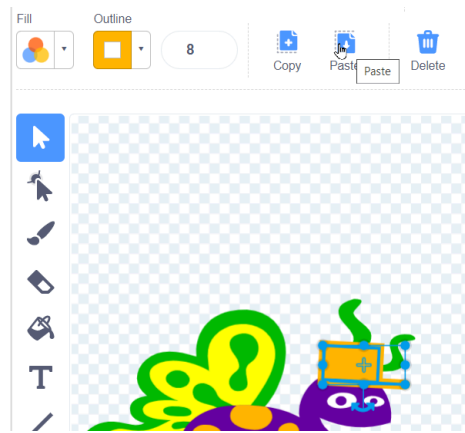
Step 4



Step 5



Step 6



Mini-Challenge 3: Set Graphic Effects

Several kinds of graphic effects can be used on your sprite using the *change [color] effect block*. These include color, fisheye, whirl, pixelate, mosaic, brightness, and ghost. You can set the effect to a number between 1 and 100, or you can change an effect a little at a time. To eliminate effects after applying them, use the *clear graphic effects* code block. Along with effects, you can set or change the sprite's size.

Procedure

1. Above the forever loop, add a *set size to (___) %* block.
2. Above the forever loop, add a *clear graphic effects* block.

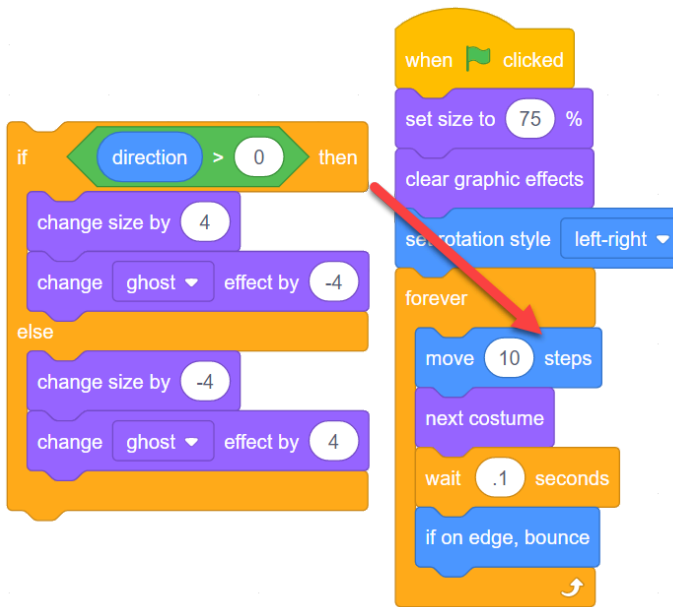
Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BUTTERFLY sprite.
Looks	set size to (75) %		
Looks	clear graphic effects		This is a section of the code stack <i>above</i> the forever loop.
Motion	set rotation style [left-right]		
Control	forever loop		The new code is RED .

Mini-Challenge 4: Change Graphic Effects

If you place the *change graphic effects* code block in a loop, the change is gradual. If it is placed inside the *forever loop*, eventually some effects will distort the sprite or view so that the game won't work anymore. In this mini-challenge, we will change the size and use the ghost effect. To prevent the effect from getting out of control, we will make the sprite grow and become more visible when the sprite is moving right. We will reverse the change when the sprite is moving left. We can do this by using an *if / then* block that checks the direction the sprite faces.

Procedure

1. Add an *if/then... else* block inside the *forever loop* but above a *move (___) steps* block.
2. Add an **equals operator** and test if the sprite's direction is > 0 .
3. In the *if, then* branch, add two blocks.
 - a. change size by (5)% (one possibility)
 - b. change [ghost] effect by (-4) (one possibility)
4. In the *...else* branch, add two blocks.
 - a. change size by (-5)% (This is the negative of the number in the *if/ then* branch.)
 - b. change [ghost] effect by (4) (The negative of the number in the *if/ then* branch.)

Menu	Code	Code block	Notes
Control Motion Operators	if / then < direction > 0 >		Put code stack on the BUTTERFLY sprite.
Looks	change size by		
Looks	change ghost effect		
Looks	change size by		
Looks	change ghost effect		

Full Script

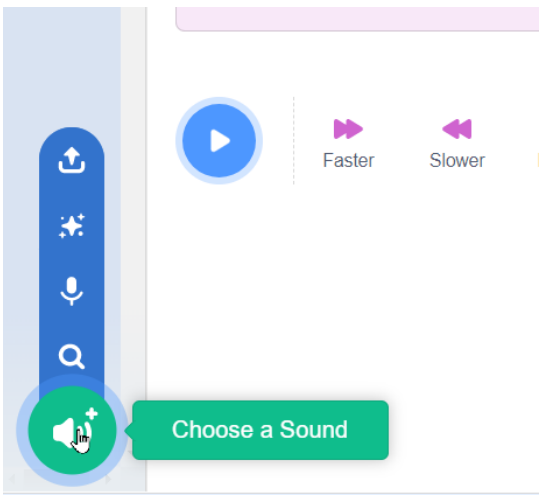
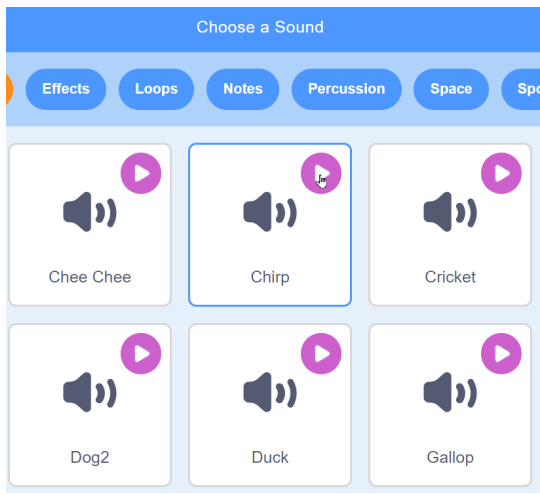
Events	when flag clicked	
Looks	set size to 75%	
Looks	clear graphic effects	
Motion	set rotation style	
Control	forever loop	
Control Motion/ Operators	if/ then ... else loop < direction > 0 >	
Looks	change size by (4)	
Looks	change ghost effect by (-4)	
	else	
Looks	change size by (-4)	
Looks	change ghost effect by (4)	
Motion	move 10 steps	
Looks	next costume	
Control	wait (.1) seconds	
Motion	if on edge, bounce	

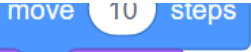
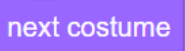
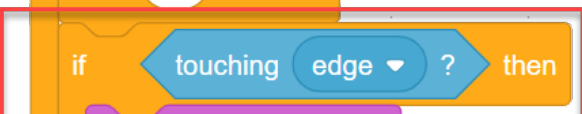
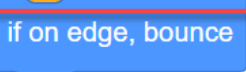
Mini-Challenge 5: Use a Sound

Each sprite comes with a default sound - usually “Pop.” If you open the sounds tab, you’ll see it. This is a short sound and is often used to enhance an event. If you want a different sound, you can add a sound from the “Choose a Sound” menu by clicking on the speaker button. You can preview sounds by hovering over the play button in the listing.

Procedure

1. Select a short sound to add to your stack from the “Choose a Sound” menu.
2. Add an *if/then* block inside the loop, just before the *if on edge, bounce* block.
3. Drop a Sensing “touching (edge)” diamond into the *if/then* code block diamond, and set it to *touching edge*.
4. Add a *start sound* block, and choose the sound you chose in step 1 using the dropdown arrow.

Step 1	
	

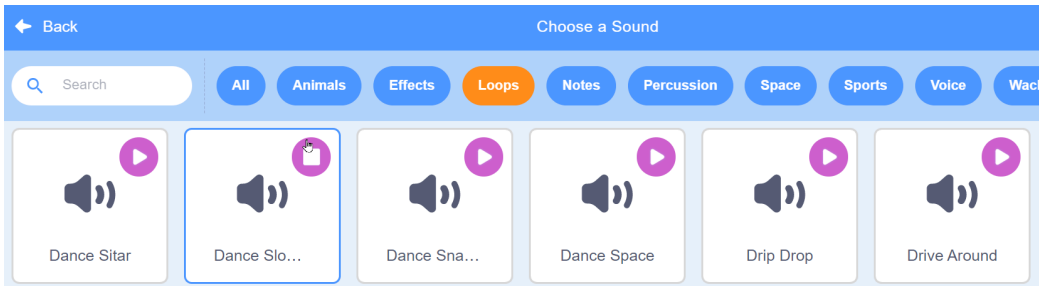
Menu	Code	Code block	Notes
Motion	move 10 steps		Put code stack on the BUTTERFLY sprite.
Looks	next costume		
Control	wait (.1) seconds		This is a section of the code stack <i>INSIDE</i> the if /then loop.
Control Sensing	if <__> then < touching (edge) >		
Sounds	start sound		
Motion	if on edge, bounce		The new code is RED .

Mini-Challenge 6: Use a Loop Sound

A loop sound is a special sound that can be repeated to sound like background music. If you want to use a loop sound this way, you create a special code stack just for the loop. Instead of using the *start sound* block, you use the *play sound until done* block. This pauses the code stack until the sound is done.

Procedure

- 1. Choose a sound from the Loops category of the Choose a Sound menu.



- 2. Start a new code stack with a *when flag clicked* block.
- 3. Add a *forever loop*.
- 4. Inside the forever loop, add a *play sound until done* block.
- 5. Select the sound loop you loaded in Mini-Challenge 5.

Menu	Code	Code block	Notes
Events Control Sound	When Clicked Forever loop Play sound until done		Put code stack on the BUTTERFLY sprite. This is a second code stack.

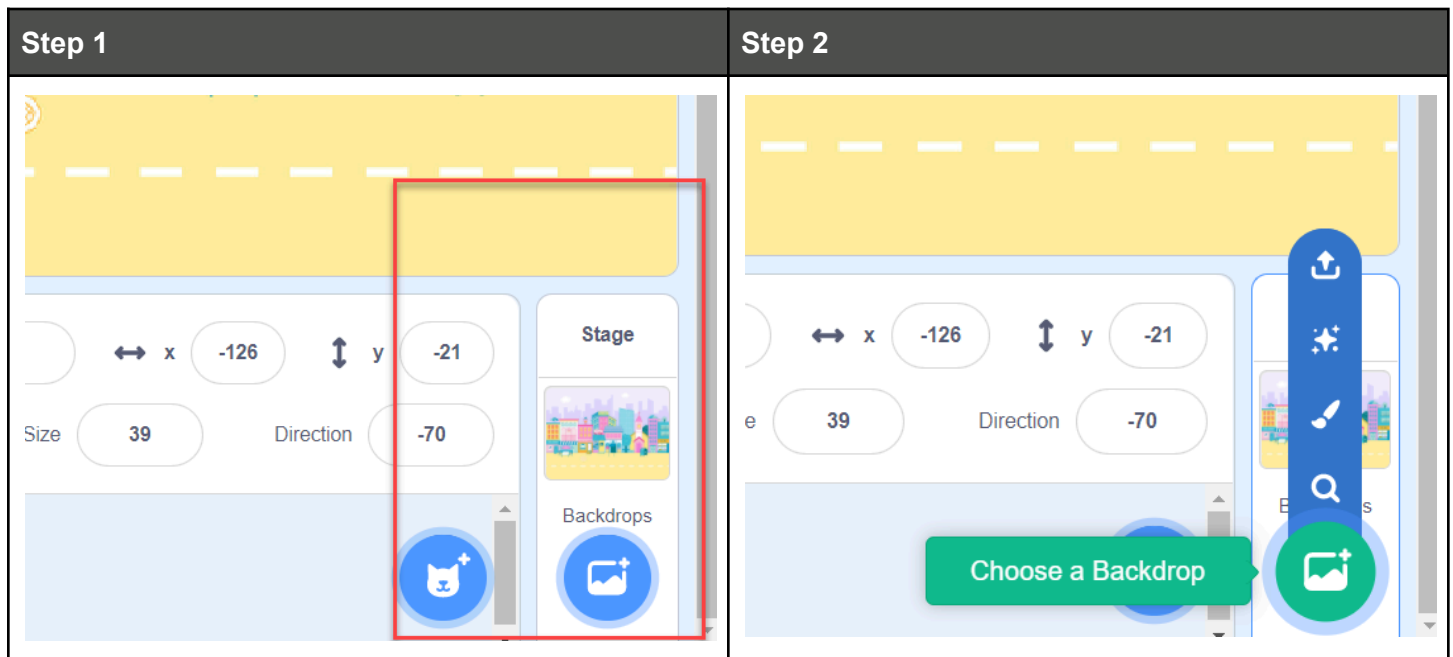
Mini-Challenge 7: Create a Background Change

You can add more backdrops. These can be used to show a change in the game, such as going from outside to inside. They can also be used to indicate a new phase of a game, like a new level or Game Over.

In this challenge, we will move all of the code in the sprite's main *forever loop* to a *repeat (50) times* loop. Then we will move the *repeat (50) times loop* back into the *forever loop*. Then we will add a block to change the backdrop.

Procedure

1. Switch to the Stage menu by clicking on the Stage panel (see below the stage).
2. Add a new backdrop that is very different from the original backdrop.



3. Switch back to the sprite's code page. Put a *repeat (50) times* loop on the canvas.
4. Drag the code from the *forever loop* into the *repeat (50) times* loop.
5. Drag the *repeat (50) times* loop **back into the forever loop**.
6. Add a *next backdrop* block *inside the forever loop*, just under the end of the *repeat (50) times* loop.

Steps 4 & 5	Step 6

Menu	Code	Code block	Notes
Looks	Next Backdrop		This code stack is on the BUTTERFLY. Add this block below the second if / then...else loop inside the forever loop.

Mini-Challenge 9: Change Soundtrack with Background

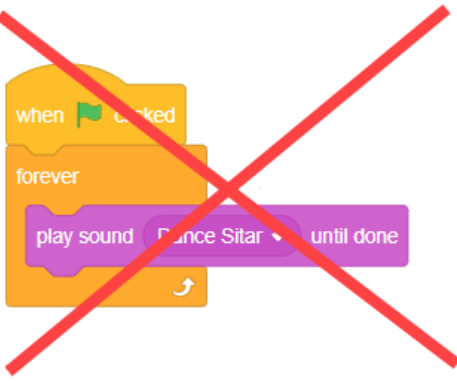
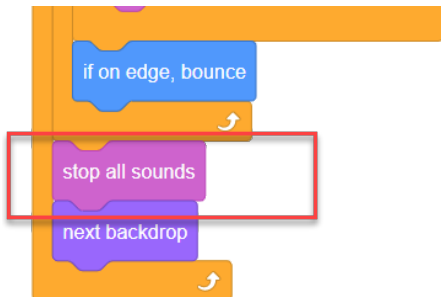
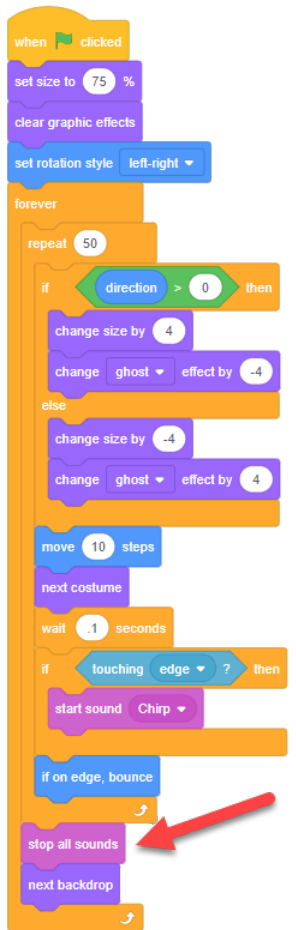
When you change the backdrop, you can enhance the mood with a sound loop that matches the mood of the new scene.

Procedure

1. Switch to the Stage menu by clicking on the Stage panel. You will get a blank code page. Don't panic! Your other code is on the sprite's code page.
2. Add two loop sounds:
 - a. One should go with the first backdrop.
 - b. One should go with the second backdrop.
3. Add a *when flag clicked* block.
4. Add a *switch backdrop* code block from the Looks category to the backdrop that you want to be first.
5. Add a *forever loop*.
6. Inside the *forever loop*, add an *if / then...else* block.
7. Drop an equal sign operator into the diamond.
8. On the left side of the equal sign operator, put a *backdrop of stage* Sensing block.
9. On the right side, enter the number 1.
10. Drop a *play sound until done* block in the *if / then* and *else* branches.
 - a. Set the first to the loop that matches backdrop 1.
 - b. Set the second to the loop that matches backdrop 2.

Menu	Code	Code block	Notes
Events	when clicked	when clicked	This code is on the STAGE .
Looks	switch backdrop to []	switch backdrop to Colorful City	
Control	forever loop	forever	Sound for backdrop 1.
Control Ops /Sensing	if / then <[backdrop #] (stage) = 1 >	if backdrop # of Stage = 1 then	
Sounds	play sound [your choice] until done	play sound Dance Sitar until done	Sound for backdrop 2.
Sounds	else play sound [your choice] until done	else play sound Dance Slow Mo until done	

11. Switch to the sprite Code page and disconnect the code block that plays the loop. (*This code is now in the Stage code stack.*)
12. Add a *stop all sounds* block just before the *next backdrop* block inside the *forever loop*.

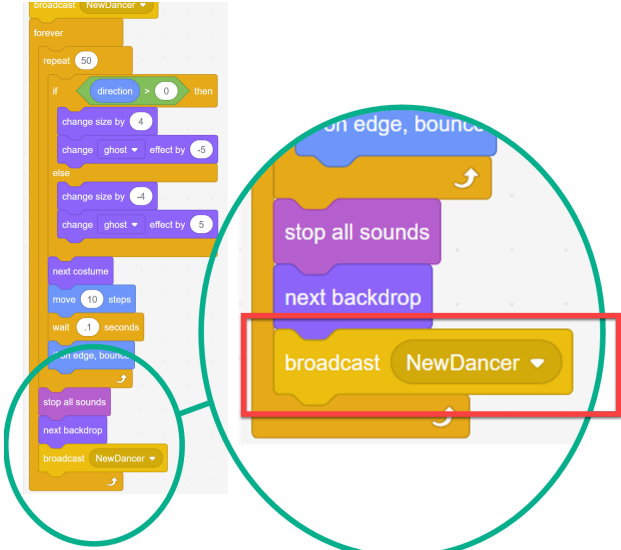
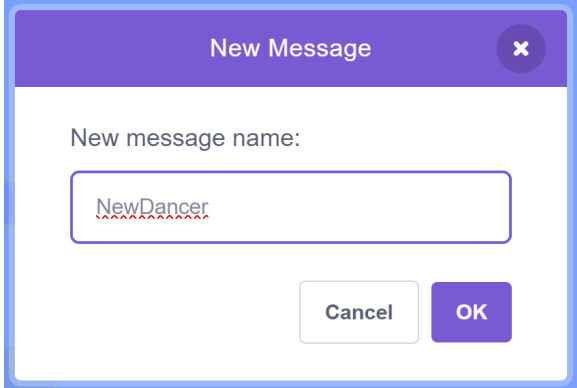
Menu	Code	Code block	Notes
			This code is on the BUTTERFLY sprite. This code stack is no longer needed.
Sounds	Stop all Sounds		

Mini-Challenge 10: Pro Tip Sprite Communication

Sprites can communicate with each other by broadcasting and receiving messages.

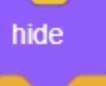
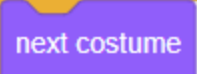
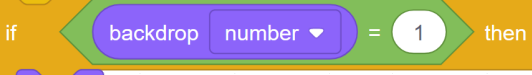
Procedure

- 1. In the main sprite's code stack, add a Broadcast Message at the bottom of the forever loop.

Menu	Code	Code block	Notes
... Sounds Looks Events	Stop all sounds Next backdrop Broadcast (new message) Click on New Message and enter the message name: NewDancer (Never use spaces OR symbols in your name.)	 	This code is on the MAIN SPRITE. The broadcast message block lets sprites send a signal to each other. It goes below the next backdrop block inside the forever loop.

Procedure

2. Add a new sprite with a dancer animation.
3. Add a *when flag clicked* block and a *hide* block.
4. Add a *forever loop* with a *next costume* and a small *wait* (0.2).
5. Start a new code stack with a *when I receive message* hat block.
6. Add an *if / then...else* block and test for the backdrop number.

Menu	Code	Code block	Notes
Events	when clicked		This code is on the DANCER sprite
Looks	hide		
Control	forever loop		
Looks	next costume		
Control	wait .2 seconds		
Events	when I receive [NewDancer]		This code is on the DANCER sprite.
Control	if < (backdrop [number]) = 1		
Looks	show		
Looks	else hide		

Procedure

7. Add a different sprite as a DANCER for the other backdrop.
8. Copy both code stacks from the first DANCER to the new sprite.
9. Change the *if / then* block to test for backdrop = 2.
10. Now you should have 2 different DANCERS appearing at different times.